

Password-less SSH Login

A guide for secure and convenient login using SSH key pairs

1. Goal

Enable SSH login **without typing a password**, using:

- SSH key authentication (public/private key)
- Optional SSH config file for convenience

2. Step 1: Generate SSH Key Pair

Run on **your local machine**:

```
ssh-keygen -t ed25519 -f ~/.ssh/my_key -C "your_email@example.com"
```

- You can change file name instead of my_key.
- Make paraphrase only when you need more security.

Press Enter to accept default file: `~/.ssh/id_rsa`

Creates:

~/.ssh/my_key (private key)

~/.ssh/my_key.pub (public key)

3. 📁📤 Step 2: Copy Public Key to Server

Option 1: Use `ssh-copy-id` (recommended)

`server_ip` is the IP address of your server.

```
ssh-copy-id -i ~/.ssh/my_key username@server_ip
```

Option 2: Manually copy (not recommended):

```
cat ~/.ssh/id_rsa.pub | ssh username@server_ip 'mkdir -p ~/.ssh && cat >> ~/.ssh/authorized_keys'
```

4. Step 3: Server-side SSH Config

Edit `/etc/ssh/sshd_config` on the **remote server**:

```
PubkeyAuthentication yes
AuthorizedKeysFile .ssh/authorized_keys
PasswordAuthentication no    # optional (disables password login)
```

Then restart SSH service:

```
sudo systemctl restart sshd
```

5. Step 4: Optional – SSH Config (Local)

Add entry in `~/.ssh/config` for convenience:

```
Host myserver
  HostName server_ip
  User username
  IdentityFile ~/.ssh/id_rsa
```

Now you can simply:

```
ssh myserver
```

 **Done!**

You can now log in **without a password**:

- No need to enter `username@host` every time
- No password prompt if private key is loaded

Questions?

Let me know if you want multi-server setup, or GitHub/CI integration!